Underwater Salient Object Detection via Image Enhancement

A comparative evaluation of preprocessing strategies for bridging the terrestrial to underwater domain gap

PURPOSE

Underwater Salient Object Detection (USOD) faces several challenges that terrestrial models don't expect. Wavelength dependent light attenuation, reduced contrast and blurred objects.

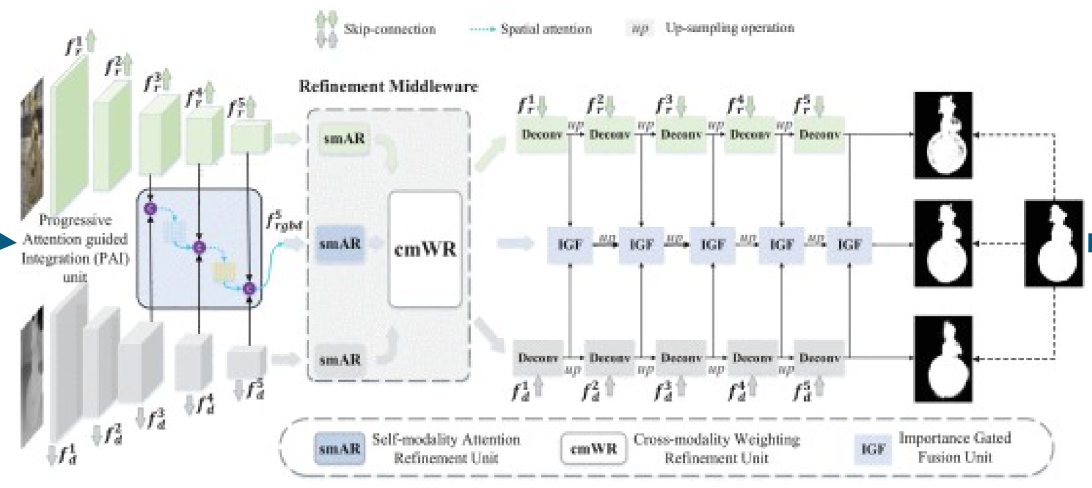
If a competent image enhancement method was used before training, inference or both, could a terrestrial SOD model compete with a USOD one?

PIPELINE

USOD10K Image

WWPF Enhancement

CIRNet SOD Model Trained on WWPF

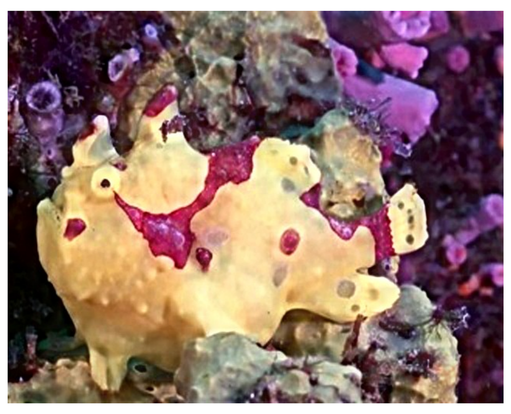


Predicted Saliency Map

ENHANCEMENT COMPARISON

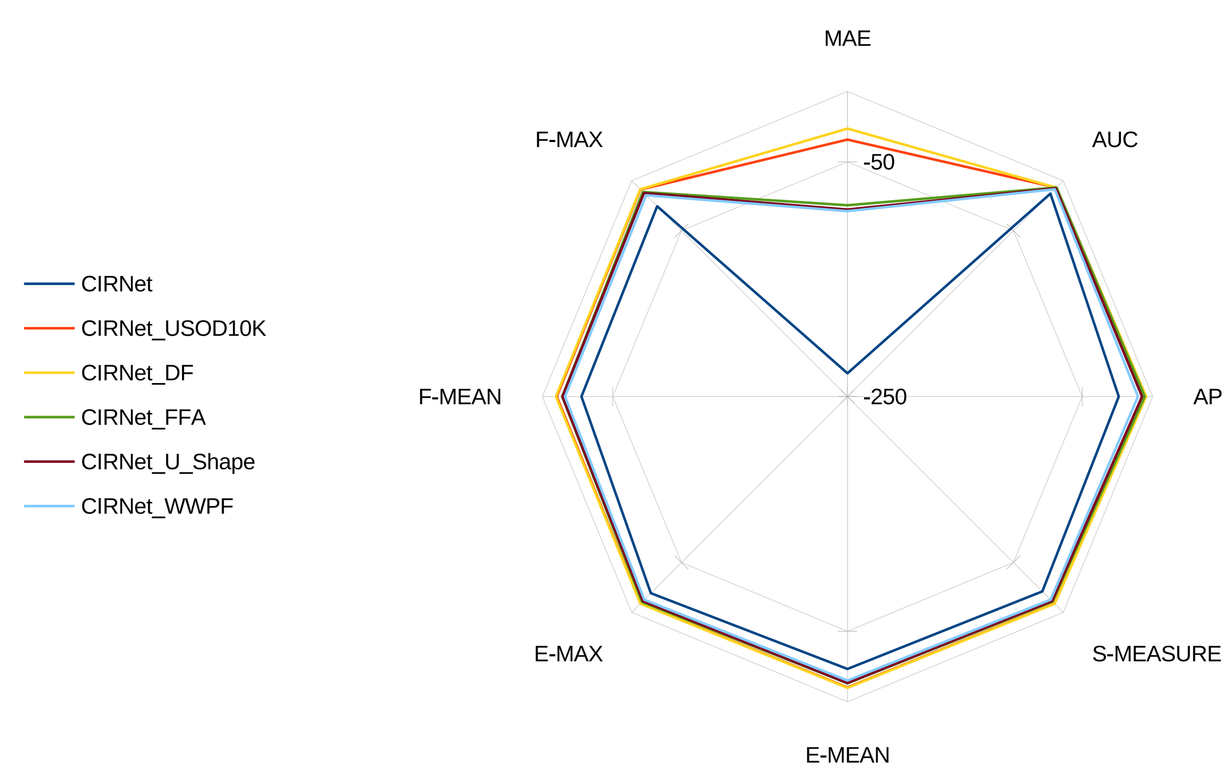
RAW

DehazeFormer Best SOD Performance



WWPF Best UCIQE Performance

SOD PERFORMANCE COMPARISON



KEY RESULTS

< 0.3% DIFFERENCE
F-MAX between CIRNet on DehazeFormer Images and TC-USOD

0 CORRELATION
Between perceptual quality metrics and SOD performance

20% FASTER
TC-USOD inference on enhanced images

CONCLUSIONS

- DehazeFormer, trained only on terrestrial haze, outperformed underwater-specific methods in SOD
- Enhancement methods that have a strong effect on the image's distribution only improve SOD when trained on the enhanced images. Otherwise they actively degrade performance.
- Image quality metrics such as UCIQE and UIQM show no reliable correlation with downstream SOD performance
- CIRNet retrained on underwater data with DehazeFormer came within 0.3% of TC-USOD on F-MAX. This means that CNN-based SOD models remain viable in certain applications.
- Architecture remains the primary performance determinant, but preprocessing offers a practical lightweight improvement for deployment-constrained scenarios

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